

IGNEOUS ROCKS

Rocks that solidify from a molten state are called igneous rocks and are broadly divided into extrusive rocks and intrusive rocks. Extrusive rocks form from lava on Earth's surface, while intrusive rocks form underground, from magma. Lava and magma are rich in silica and metallic elements. As they cool, minerals such as feldspars, micas, amphiboles, and pyroxenes form. Different combinations of these minerals make up many of the igneous rocks.

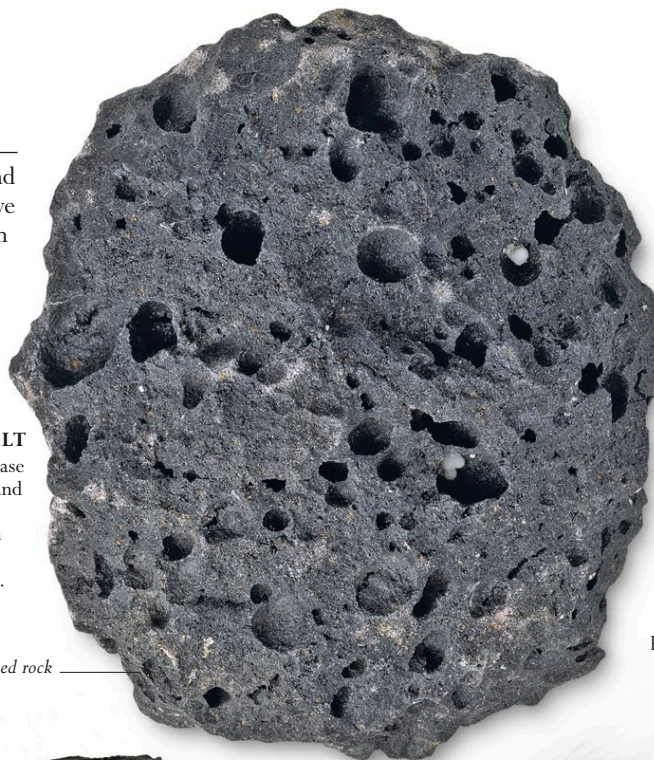
BASALT

A dark, fine-grained volcanic rock, basalt is the most common rock forming the oceanic crust.



VESICULAR BASALT

Dominated by plagioclase feldspars, pyroxenes, and olivine, this lava is dark-coloured, with numerous vesicles (former gas cavities).



dark, fine-grained rock

RHYOLITE

This fine-grained, pale-coloured lava contains much quartz, mica, and feldspar. It frequently includes visible phenocrysts (larger crystals).



vesicle
(gas bubble cavity)

BANDED RHYOLITE

Similar in composition to granite, rhyolite also has tiny crystals because it forms from rapidly cooled lava. Banding shows the direction of lava flow.



AMYGDALOIDAL BASALT

Amygdaloidal means almond-shaped and refers to the gas bubble cavities found in basaltic lavas, which are typically infilled with secondary minerals such as zeolites, carbonates, and agate.

PAHOEHOE

Common in Hawaii, pahoehoe is named after the Hawaiian word *hoe*, meaning "to swirl". This basaltic lava is also known as ropy lava.



PORPHYRITIC BASALT

This dark-coloured rock has large crystals, typically of olivine or plagioclase feldspars, set in a fine-grained matrix.



PÉLÉ'S HAIR

Named after a Hawaiian goddess, and composed of numerous slender, brown, glassy strands, Pélé's hair is formed by lava sprays being blown in the wind.



IGNIMBRITE

A very fine-grained, pale-coloured volcanic tuff, ignimbrite often shows banding caused by the flow of lava when molten.



PUMICE

Formed from frothy lava, pumice contains small crystals of feldspar in a very fine-grained matrix. It is of such low density that it floats on water.

LITHIC TUFF

This rock contains fragments of previously formed rock in a very fine-grained matrix. Usually pale, lithic tuff is formed by violent volcanic eruptions.



SPINDLE BOMB

Molten, low-viscosity basaltic lava blasted through the air can assume aerodynamic forms such as this one. These masses cool to form "bombs".

BREADCRUST BOMB

This variety of volcanic bomb is characterized by a cracked crust, due to continued expansion of the interior after the exterior has solidified.

